

Lifestyle & Health: Hormones

Quick reference sheet · Biology · keep handy for revision

1. The endocrine system

The **endocrine system** is made up of glands that secrete **hormones** directly into the bloodstream. A hormone is a chemical messenger that travels in the blood to a **target organ**, where it produces an effect. Key glands: pituitary (the "master gland"), pancreas (insulin & glucagon), adrenal glands (adrenaline), ovaries/testes (reproductive hormones).

2. Hormones vs nerves

Feature	Nervous system	Hormonal system
Signal	Electrical impulses along neurons	Chemicals (hormones) in the blood
Speed	Very fast	Slower
Effect	Short-lived, precise area	Longer-lasting, wide area

3. Controlling blood glucose concentration

Blood glucose is monitored and controlled by the **pancreas**. After eating carbohydrate, blood glucose rises → the pancreas releases **insulin** → insulin causes cells (liver/muscle) to take up glucose, converting excess into **glycogen** for storage → blood glucose falls back to normal. (Beyond Foundation: when blood glucose is too low, the pancreas releases **glucagon**, which converts glycogen back into glucose — together these keep blood glucose steady by negative feedback.)

4. Type 1 & Type 2 diabetes

	Type 1	Type 2
Cause	Pancreas produces little/no insulin	Cells stop responding properly to insulin
Risk factors	Not strongly linked to lifestyle	Obesity, poor diet, lack of exercise
Treatment	Insulin injections; monitor blood glucose	Lifestyle changes; medication; insulin if needed

Lifestyle link: obesity, poor diet and inactivity increase the risk of Type 2 diabetes — the same lifestyle changes that reduce cardiovascular disease risk also reduce this risk.